

Population-Scope Tests 51–55

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This work derives from and formalizes the inter-Self coordination architecture introduced in “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026), the third paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026) and “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026).

Abstract

This note formalizes five binary operational tests — Tests 51 through 55 — for governance properties relevant to CKS-governed Selves operating at population scope under Claim 6 of the third CKS theory paper. The tests are written in the calibrated-humility register: they surface governance properties that are possible and recommended at population scale, but whose absence does not constitute a violation of the architectural commitments that make a Self CKS-coherent at Claims 1–5 scope. Accordingly, a FAIL on any of Tests 51–55 is less architecturally severe than a FAIL on a Claims 1–5 compliance test. The partial exception is Test 55, whose FAIL identifies AP-27 (Forced Propagation) — an anti-pattern that is architecturally prohibited regardless of scope. Each test specifies records to examine, a binary question, and what PASS and FAIL indicate.

1. Register framing: population-scope tests and calibrated humility

Tests 51–55 occupy a different severity register than the compliance tests that verify Claims 1–5. The Claims 1–5 compliance tests ask whether a Self’s substrate, FAI events, conflict handling, evolution feed, and governance configuration satisfy the architectural commitments that constitute CKS-coherence. A FAIL on a Claims 1–5 compliance test indicates that the Self is operating outside the architectural pattern. The correction required is architectural.

Tests 51–55 ask different questions. They ask whether a Self that is CKS-coherent at Claims 1–5 scope has also authored the governance content that makes its population-scope participation well-specified, capacity-bounded, current, transparent, and free of coercive relationship structures. These are properties that good governance at population scope makes possible. They are not

properties that any single FAI event, or any home-substrate operation, requires. A Self may be fully CKS-coherent — its substrate governed, its FAI events aspect-level merges under joint human authority, its conflicts preserved, its evolution feed routed correctly, its configuration recursive — while having a participation configuration that is incomplete, capacity-unconstrained, stale, non-transparent, or (in the most severe case) coercively structured.

The FAIL condition for Tests 51–54 is best read as a governance gap: governance content that would make the Self’s population-scope posture explicit has not been authored. The gap is advisory. It should be addressed; it does not make the Self architecturally non-compliant with the CKS pattern. This is the calibrated-humility register the note applies throughout.

Test 55’s FAIL is in a different class, addressed separately in Section 7.

2. Severity hierarchy across the five tests

Before the five tests are stated, naming the severity ordering explicitly avoids misreading any FAIL as equivalent:

Tests 51, 52, 53, 54 — advisory governance gaps. FAIL indicates missing or incomplete governance content. The Self’s population-scope posture is partially implicit. The consequence is governance ambiguity, not architectural incoherence. Correction is authoring the missing content.

Test 55 — prohibited anti-pattern (AP-27). FAIL indicates that an absorption obligation provision is present in a governance arrangement, compromising home governance sovereignty over DNA absorption. This is architecturally prohibited regardless of scope. Correction requires removing the offending provision, not merely filling a governance gap.

The remaining sections state each test in the standard format.

3. Test 51 — Participation Configuration Completeness

Source commitment: D2.31, Claim 6.

Records to examine: Network participation configuration for the Self being audited.

Binary question: Does the Self’s network participation configuration specify all five dimensions — (A) propagation permissions, (B) propagation restrictions, (C) partner preferences, (D) network role, and (E) review cadence?

PASS indicates: All five dimensions are explicitly authored as governance content. The Self’s population-scope posture is fully specified: what it will contribute to shared substrates, what it will decline to contribute, the partner characteristics it will favor or avoid, the role it expects to play in FAI events (initiator, responder, both), and how frequently governance will review and update the configuration. A partner Self encountering this configuration can assess the relationship’s

compatibility with its own posture without inferring anything from behavior.

FAIL indicates: The participation configuration is absent, or one or more of the five dimensions is not explicitly authored. The Self's network participation posture is partially implicit. The governance ambiguity this creates is bilateral: the Self's governance team cannot confirm that participation intent has been fully articulated, and prospective partner Selves cannot assess compatibility without observing the Self's behavior across prior FAI events — which is a weaker and slower signal than explicit configuration. The FAIL does not indicate that the Self's prior or current FAI events were conducted incorrectly. It indicates that population-scope posture has not been fully reduced to governance content, which is the appropriate condition for explicit governance at this scope.

Calibrated-humility register note: A participation configuration that specifies only some dimensions is a governance gap, not an architectural failure. The test identifies what remains to be authored.

4. Test 52 — Governance Capacity Ceiling in Participation Configuration

Source commitments: D2.33, D4.07.

Records to examine: Network participation configuration; governance capacity assessment records.

Binary question: Does the participation configuration reference a governance capacity ceiling — the maximum number of simultaneous FAI events governance can manage — and specify how additional invitations are handled when capacity is reached?

PASS indicates: The capacity ceiling is explicitly specified as governance content within the participation configuration, and the configuration specifies what happens when capacity is reached: invitations above the ceiling are deferred, declined, queued with acknowledged delay, or handled by some other explicitly described mechanism. The configuration connects the Self's network participation posture to actual governance capacity, preventing the participation configuration from creating an invitation funnel that governance cannot service. FAI events require human governance attention — joint authority over the shared substrate, configuration of ingestion, conflict escalation across perimeters. A Self that accepts FAI invitations without bound will eventually accept more than governance can attentively manage at any given time.

FAIL indicates: The capacity ceiling is absent from the participation configuration. The configuration may be complete across all five dimensions of Test 51 and still fail Test 52 if it does not bound the simultaneous-event load governance can carry. The consequence is an unconstrained invitation funnel: the Self's stated propagation permissions and partner preferences will generate invitations that governance has not bounded, and overflow events — events accepted when governance is already at capacity — may receive less than full governance attention. The

FAIL is advisory. It indicates that the participation configuration has not been aligned with governance capacity assessment; it does not indicate that any specific FAI event was conducted with deficient governance.

Calibrated-humility register note: Governance capacity is an empirical property of the Self's governance team and review cadence. Specifying a ceiling requires governance to assess its own capacity, which is itself a governance act this test encourages rather than mandates.

5. Test 53 — Participation Configuration Currency

Source commitments: D2.61, D4.18.

Records to examine: Participation configuration; exit records for any exited relationships; standing configuration revocation records.

Binary question: Does the participation configuration accurately reflect the Self's current FAI relationship portfolio — with exited relationships removed, standing configurations revoked, and partner preferences reflecting current governance intent?

PASS indicates: The participation configuration is current. Exited relationships have been removed. Standing configurations for relationships that have been terminated have been revoked. Partner preferences reflect the governance team's current assessment of the population-scope relationships the Self should pursue. The configuration is a reliable signal of present posture, not a historical archive of past posture.

FAIL indicates: The participation configuration references exited relationships, retains standing configurations for terminated relationships, or carries partner preferences that reflect a prior governance intent no longer operative. The configuration is a stale network topology signal; it may generate invitations to partners with whom the Self no longer maintains a relationship, or it may present a posture to partner Selves that governance has superseded.

There is a specific and important connection between Test 53 FAIL and AP-13 (Silent Offboarding) at network level. AP-13 names the anti-pattern in which a Self exits a relationship without completing the governance record of that exit — the offboarding occurred, but the governance residue of the relationship persists in the substrate without marking the exit explicitly. When a Self's participation configuration still references a relationship after that relationship has been exited, Test 53 detects the network-level form of this residue: the offboarding was completed at the relationship level but the participation configuration was not updated. The stale configuration entry is not evidence that the exit was wrongly handled; it is evidence that exit governance did not include a configuration-update step. Test 53 creates the operational test that would catch this omission at audit time, and thereby motivates governance practices that include configuration update as a required step in relationship exit workflows.

Calibrated-humility register note: Configuration currency is a maintenance property, not a structural one. A newly authored, accurate configuration that becomes stale over time due to relationship exits that were not propagated to the configuration is a governance maintenance gap, not an architectural failure.

6. Test 54 — Participation Mode Transparency

Source commitment: D2.62.

Records to examine: Network participation configuration (propagation permissions Dimension A and restrictions Dimension B); FAI event history (contributions versus absorptions).

Binary question: If the Self is operating in a non-full participation mode — contribute-only, absorb-only, or restricted — does the participation configuration explicitly specify this mode and include a governance justification for the non-full mode?

PASS indicates: Either (a) the Self operates in full participation mode (contributes and absorbs in FAI events), and the participation configuration is consistent with this mode and the FAI event history confirms consistent contributions and absorptions; or (b) the Self operates in a non-full mode, the participation configuration explicitly names the mode (contribute-only, absorb-only, or restricted with specified restrictions), and the configuration includes a governance justification authored under the Self's home governance authority that explains why the non-full mode is the appropriate posture. A governance justification is not a technical argument; it is a governance-team statement that names the reason the mode was chosen — strategic content protection, capacity constraints, relationship-specific conditions, or other explicitly stated considerations.

FAIL indicates: The participation mode is absent from the configuration, or a non-full mode is operating without a governance justification. The consequence is that participation mode gaming — AP-26, in which a Self systematically absorbs governance content from partner Selves while contributing minimally or nothing in return, without disclosing this posture — becomes possible and undetectable through configuration inspection. Partner Selves cannot assess a Self's contribution posture without observing its behavior across many events. An explicit mode specification with justification reduces the information asymmetry that makes AP-26 possible. The absence of this specification does not confirm that AP-26 is occurring; it removes a governance signal that would make the posture transparent.

Calibrated-humility register note: Non-full participation modes can be entirely legitimate — a Self with highly specialized content may contribute-only; a newly formed Self may absorb-only during an initial accumulation period. The test does not penalize non-full modes; it tests whether the mode is governed transparently.

7. Test 55 — Forced Propagation Absence

Source commitments: D2.32, AP-27.

Records to examine: Cross-organizational agreements; standing configurations; FAI event configurations.

Binary question: Do any cross-organizational agreements, standing configurations, or FAI event configurations contain provisions requiring absorption of specific governance patterns as a condition of participation or continued relationship?

PASS indicates: No absorption obligation provisions are present in any governance arrangement — agreements, standing configurations, or FAI event configurations. A Self’s decision to absorb any content from a shared substrate into its home substrate remains at all times a governance decision made under home governance authority, consistent with Claim 5’s recursive applicability and Paper 1’s human-governed commitment. Partner Selves may offer content; they may communicate the value of particular patterns; they may structure FAI events to facilitate absorption. What they may not do, within a CKS-governed network, is make absorption obligatory as a condition of participation.

FAIL indicates: An absorption obligation provision has been found in a governance arrangement. AP-27 (Forced Propagation) is present. Home governance sovereignty over DNA absorption has been compromised: the Self’s governance team can no longer make a fully independent decision about what to absorb from a shared substrate, because a prior agreement or standing configuration has already committed the Self to absorb specific content as a condition of an ongoing relationship.

Severity note: Test 55 FAIL is architecturally more severe than Tests 51–54 FAIL, and the calibrated-humility register applies with a different force here. Tests 51–54 identify governance gaps — content that should be authored has not been. Test 55 identifies an anti-pattern: content that should not be present is present. The distinction is between absence and commission. AP-27 is architecturally prohibited regardless of whether the Self is CKS-coherent at Claims 1–5 scope in all other respects. A Self that otherwise satisfies all Claims 1–5 compliance tests but carries forced-propagation provisions in its governance arrangements has not merely left governance content unwritten; it has structured a governance arrangement that constrains home governance authority in a way the CKS pattern does not permit at any scope.

The correction for Test 55 FAIL is not authoring missing content. It is removing the offending provision and, where the provision was embedded in a cross-organizational agreement, renegotiating the agreement to remove the absorption obligation. This may require unwinding a relationship structure, which is a more demanding correction than filling a governance gap.

8. Applying the five tests together

The five tests are independent: a Self can PASS any subset and FAIL the remainder. In practice, certain FAIL combinations are more informative than individual failures.

A Self that FAILs Test 51 (configuration incomplete) and Test 53 (configuration stale) simultaneously may have a participation configuration that was partially authored at the time of some earlier relationship portfolio and then not updated. The combined FAIL indicates that both the configuration's coverage and its currency need governance attention.

A Self that FAILs Test 54 (mode unjustified) and whose FAI event history shows a consistent absorb-heavy pattern without contribution should be examined for AP-26 risk. The FAIL does not confirm the anti-pattern, but it removes the governance transparency that would allow the pattern to be distinguished from legitimate absorb-oriented operation.

A Test 55 FAIL, discovered in combination with any Tests 51–54 FAIL, should be prioritized: AP-27 requires correction before governance attention returns to the advisory gaps Tests 51–54 identify.

9. Conclusion

Tests 51–55 extend the operational test vocabulary into population scope under Claim 6. They ask whether a CKS-coherent Self has authored the governance content that makes its network participation posture explicit, capacity-bounded, current, transparent, and free of coercive structure. The calibrated-humility register holds throughout: Tests 51–54 FAILs identify governance gaps at population scope, less severe than Claims 1–5 compliance failures, remedied by authoring the missing content. Test 55 FAIL identifies AP-27, an architecturally prohibited anti-pattern remedied by removing the offending provision, not by authoring missing content. The severity distinction matters for audit prioritization: Tests 51–54 FAILs inform the governance team about work to be done at population scope; Test 55 FAIL informs the governance team that a structural correction is required immediately.

Population-scale CKS-governed Self deployments do not yet exist. These tests are specifications for governance audits that become operational as the architecture's population-scope extension matures. They are anticipated governance instruments, written in advance of the operational context they will serve, consistent with the calibrated-humility voice anchor that the trilogy's extension claim registers require.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model*. April 2026. ORCID: 0009-0004-8065-3235.

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